

Chapter 120 — Aging and Geriatric Urology

- **Urology is a predominantly geriatric specialty:** geriatricians aside, only ophthalmology and cardiology exceed it in geriatric care volume. Geriatric competencies are recommended for surgical training and board certification.
- **Age raises need and risk together:** more need for care, more risk from surgery and medication. **Age alone matters less than** geriatric syndromes, frailty, function, mental status and socioeconomic factors.
- **Aim:** mitigate risk factors, preserve freedom from frailty and vulnerability, minimise intervention risk. This includes **altering goals of care** in ways inappropriate for a younger patient with the same condition.
- **Ignatz Leo Nascher (1909)** first used *geriatrics* as a clinical term; it is now an established specialty.
- **Conventionally ≥ 65 , but chronological age is a poor descriptor:** "elderly" starts anywhere from 50 to 80 in the hip-fracture literature. The United Nations proposes defining it by **years of life remaining**, to reflect variation in life expectancy and access to care.
- **No typical 65-year-old:** environment, health behaviours, family, gender and ethnicity shape ageing. **UN Decade of Healthy Ageing 2021–2030**, led by the World Health Organization (WHO).

Concept — *The chapter replaces chronological age with **physiological and functional age**. Function, frailty and geriatric syndromes are the tools for measuring it.*

DEMOGRAPHICS OF AGING

Aging and Population Trends

- **Adults ≥ 65 are the fastest-growing US segment:** they will nearly double to 94.7 million between 2016 and 2060.
- **By 2034, people ≥ 65 will outnumber children** for the first time in US history.
- **People ≥ 85 will triple**, from 6 to 19 million, by 2060.
- Drivers: **longer lives, falling birth rates and medical advances**.
- *Figure 120.1* — US population pyramids for 1980, 2022 and the 2060 projection (US Census Bureau; Vespa et al., 2020). The base narrows and the older bands widen as the pyramid becomes a column.

BIOLOGY AND PRINCIPLES OF AGING

- **Ageing is the accumulation of molecular and cellular insults.** It leads to declining physical and mental capacity, rising disease risk and, finally, death.
- **Decline is not linear with chronological age.** It may be gradual or rapid, which explains the wide functional diversity in old age. Environmental and modifiable factors, including life transitions such as retirement, rehousing and bereavement, strongly shape healthy ageing.
- **Cellular senescence (a cell stops replicating) is the fundamental driver of biological ageing;** senescent cells accumulate with age. Telomere-driven arrest defines the **Hayflick limit** (1961), which correlates with lifespan.
- **Senescence-associated secretory phenotype (SASP):** cytokines, growth factors and matrix metalloproteinases → tissue remodelling, tumorigenesis, chronic inflammation.
- **Senescence spreads:** SASP cells age adjacent tissue. Senescent cells transplanted from old into young mice spread senescence to the host.

REPLICATIVE VERSUS STRESS-INDUCED SENESENCE

	Replicative senescence	Stress-induced senescence
Trigger	Telomere shortening with each division	Reactive oxygen species, radiation, DNA damage, unfolded protein response
Framework	Hayflick limit; telomere theory of ageing	Telomere loss is only one route to arrest
Common result	Irreversible arrest, tumour-suppressor genes up, cell-cycle genes down, SASP	

Concept — *Senescence is both an endpoint and an engine. SASP turns the arrest of single cells into tissue-wide inflammatory ageing, so aged tissue ages the tissue around it.*

Healthy Aging

- **WHO definition:** "the process of developing and maintaining the **functional ability** that enables wellbeing in older age."
- **Functional ability** is the capability to be and do what one values. It has five parts: (1) meet basic needs; (2) learn, grow and make decisions; (3) be mobile; (4) build and maintain relationships; (5) contribute to society.
- **Functional ability comes from intrinsic capacity interacting with environment.** Intrinsic capacity is the person's mental and physical capacity; environment is home, community and society.
- Purpose: gives clinicians a **goal or target for care**.

Concept — *Healthy ageing is defined by function, not by the absence of disease. That is why the clinical evaluation that follows begins with function.*

CLINICAL EVALUATION OF THE GERIATRIC UROLOGY PATIENT

- Older adults have **more frequent and more complex health problems**. Physiological ageing is compounded by **interrelated geriatric syndromes** (for example frailty), which do not affect younger adults and act synergistically. They are more complex than simple comorbidity.
- The result is **loss of functional reserve capacity** and an impaired response to stressors such as infection, surgery, chemotherapy and urological disease or treatment.
- **Chronic disease burden:** about four in five adults over 65 have at least one chronic condition, and about one in four have four or more.
- **First step: assess the ability to function independently.**

HOMEOSTASIS VERSUS HOMEOSTENOSIS

Homeostasis	Homeostenosis
The body restores or keeps balance in response to internal or external stressors	Age-associated narrowing of functional reserve , named by analogy with atherosclerotic stenosis. Baseline reserve is lower and balance is restored less well after a stressor

TRADITIONAL VERSUS SYNERGISTIC DISEASE MODEL (FIG. 120.3; DECKER AND SAUSVILLE, 2005, MODIFIED BY INOUE, 2007)

Traditional linear model	Synergistic additive model (geriatric syndromes)
One significant risk factor leads to early disease, then advanced disease; intervention acts on that chain. Fits single-driver conditions such as inborn errors of metabolism or trauma	Multiple risk factors interact synergistically to produce the clinical phenotype. The end result may be functional decline, another geriatric syndrome or worsening of a contributor

Functional Assessment

- **Baseline functional status is vital.** It predicts remaining life expectancy, morbidity and mortality. Heterogeneity is wide, so factors other than chronological age drive disability.

Activities of Daily Living

- **Basic activities of daily living (BADLs, ADLs)** are graded from complete independence to full assistance; their loss predicts dependency on caregivers.
- **Screening question:** can the patient shower or bathe independently? This is an early sign of general decline.
- **ADLs predict postoperative outcome**, and identifying difficulty is still worthwhile when no surgery is planned. **Occupational therapy** (home or community) restores some independence, social participation and mobility.

Instrumental Activities of Daily Living

TIERS OF DAILY-LIVING FUNCTION

Tier	Content	Significance
BADLs (ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	Bathing is lost first; feeding is preserved longest
Instrumental activities of daily living (IADLs)	Cooking, cleaning, transport, laundry, finances; telephone or digital devices; managing medications	Critical to independent living; often used interchangeably with ADLs
Advanced activities of daily living (AADLs)	Named as a separate tier; content not detailed in the chapter	—

- **Graded loss of function predicts survival and cost** (Medicare Current Beneficiary Survey cohort aged 70). Full independence → longest life expectancy and lowest cost; losing one IADL → shorter and costlier; needing help with at least one BADL → shortest and costliest.
- **Self-rated "poor" or "very poor" health predicts ADL and IADL decline** over 3 years in both sexes. Mobility, cognition and depression influence function, and a baseline lets you **monitor change** over time and with treatment.

Concept — *Function is a hierarchy lost from the top down. Complex IADLs go before basic ADLs, and bathing goes before feeding. Each step down costs life expectancy and money.*

Surgical Risk and Medical Optimization

- **Frailty (impaired compensation for stressors)** can bias surgeons and patients against surgery without anyone measuring it.
- **Age by itself should not be a limiting factor**; comorbidity may matter more, and optimisation can minimise complications even over 80.
- **Recognising comorbidity and geriatric syndromes is the surgeon's responsibility**; to "clear" a patient is a misnomer.
- **Chronological age is seldom an independent predictor** of surgical morbidity or mortality. Predictors: longer operating time, poor American Society of Anesthesiologists (ASA) score, urgent or emergent priority.
- **Society for Perioperative Assessment and Quality Improvement (SPAQI) sequence**: frailty screen before elective major surgery → if positive, diagnostic frailty assessment → comprehensive geriatric assessment where feasible.

PERIOPERATIVE GUIDANCE FOR THE OLDER SURGICAL PATIENT

Guidance	Bodies	Scope
Optimal Preoperative Assessment of the Geriatric Surgical Patient (Chow et al., 2012)	American College of Surgeons (ACS) National Surgical Quality Improvement Program (NSQIP) and American Geriatrics Society (AGS)	Preoperative: 117 studies, mostly NSQIP data; targeted assessments beyond ADLs, IADLs, cognition and mobility (Box 120.1)
Optimal Perioperative Management of the Geriatric Patient (Mohanty et al., 2016)	ACS NSQIP and AGS	Extended to the whole perioperative period: interventions that improve outcome
Preoperative Management of Frailty (2018)	SPAQI	Frailty assessment as a standard part of preoperative assessment before elective major surgery

Box 120.1 Checklist for the Optimal Preoperative Assessment of the Geriatric Surgical Patient

- **Mind:** (1) cognitive ability and capacity to understand the surgery; (2) depression screen; (3) risk factors for postoperative delirium; (4) alcohol and substance misuse or dependence.
- **Organ systems:** (5) cardiac evaluation using the American College of Cardiology/American Heart Association algorithm for non-cardiac surgery; (6) risk factors for postoperative pulmonary complications, with preventive strategies.
- **Function and reserve:** (7) functional status and history of falls; (8) baseline frailty score; (9) nutritional status, with preoperative intervention if nutritional risk is severe.
- **Medicines:** (10) detailed medication history, perioperative adjustment and monitoring for polypharmacy.
- **Goals and support:** (11) treatment goals and expectations weighed against likely outcomes; (12) family and social support.
- **Tests:** (13) preoperative investigations appropriate to older patients.

From Chow et al., J Am Coll Surg 2012, Table 1.

- **Pattern and trap:** only 2 of 13 items concern organ systems (cardiac, pulmonary). The rest cover geriatric domains that routine work-up misses. It is a tool for optimisation, not clearance.

Comprehensive Geriatric Assessment

- **Comprehensive geriatric assessment (CGA):** a multidisciplinary assessment, usually led by a geriatrician. **No standardised definition yet;** it generally covers functional assessment and an inventory of the patient's geriatric syndromes.
- **Sequence:** frailty assessment (commonly the Fried criteria) → physical performance (standing balance, a timed 360° turn as fast as possible, three chair rises, picking a pen up from the floor, putting on and removing a jacket) → laboratory tests (serum albumin, creatinine, brain natriuretic peptide) → respiratory function, forced expiratory volume in 1 second (FEV₁).

Concept — Optimise, do not "clear." *Surgical risk comes from physiological reserve and operative context (ASA score, urgency, duration), not from date of birth.*

Major Geriatric Syndromes and Urology

- **Definition:** ageing-related dysfunctions **spanning multiple organ systems**. They are associated with reduced life expectancy, disability, dependent care, hospitalisation and mortality, and are often poorly understood by urologists.
- **Interrelated and synergistic:** frailty is linked to urinary incontinence (UI) and falls, and improving frailty reduces both. Falls increase frailty, and rushing to the toilet links incontinence to falls.

- **Major syndromes:** frailty, depression, delirium, falls, incontinence, polypharmacy, vision impairment and hearing impairment. They worsen cardiovascular, diabetic, pulmonary, malignant and neurological disease.
- **Lifelines Cohort Study** (Netherlands, adults ≥ 60): **one or more geriatric syndromes raise the risk of a new chronic condition**, independent of socioeconomic status, social contact, health status and smoking.
- **Balance and cognitive impairment raise the need for dependent care; chronic pain does not.**

Frailty

COMORBIDITY VERSUS DISABILITY VERSUS FRAILITY

Comorbidity	Disability	Frailty
Coexisting conditions (for example diabetes, chronic obstructive pulmonary disease), regardless of how they interact	Inability to care for oneself , often measured by ADLs and needing daily help	Decreased reserve, so increased vulnerability to stressors. Surgery is the stressor that matters most to urologists

MODELS OF FRAILITY

	Accumulated deficits model	Fried phenotype (Fried et al., 2001)
Method	Count deteriorating systems and combine them into a frailty index, for example the deficit accumulation index (DAI) ; more deficits give a higher score	Five measurable components: (1) unintentional weight loss >10 lb (about 4.5 kg) or 5% of body weight in 1 year ; (2) reduced grip strength; (3) slower gait and mobility; (4) reduced daily activity; (5) early exhaustion with activity
Cut-off	Continuous index	≥ 3 components = frail; 1-2 = pre-frail
Validity	Not predictive of postoperative outcome , probably because the interaction and severity of deficits cannot be captured by counting them	Validated over two decades: predicts death, loss of independence, falls, hospitalisation, poorer response to treatment, delayed recovery and adverse postoperative outcome

- **Prevalence rises with age:** about 7–10% of community-dwelling adults at 65, rising to 25% over 85.

Concept — *Frailty is loss of reserve, best captured as a phenotype. Counting deficits fails because what matters is how deficits interact, not how many there are.*

Polypharmacy and Medication Optimization

- **Polypharmacy** means taking multiple medicines. It is not dangerous in itself, but **with advanced age and harmful interactions it becomes a geriatric syndrome** that worsens the others.
- **Inappropriate medicines cause side effects and non-medical harm**, including financial toxicity. A key driver is the "**prescribing cascade**."

TABLE 120.1 BEERS CRITERIA OF UROLOGY-RELEVANT MEDICATIONS (REBUILT)

Group	Drug and setting	Concern	Action and alternatives
Anticholinergic burden	Urinary antimuscarinics (darifenacin, fesoterodine, flavoxate, oxybutynin, solifenacin, tolterodine, trospium) in dementia, cognitive impairment, delirium or high delirium risk	Confusion, cognitive impairment, delirium, dry mouth, constipation	Avoid combining anticholinergic drugs (cognitive decline)

Group	Drug and setting	Concern	Action and alternatives
	Tricyclic antidepressants (amitriptyline, clomipramine, doxepin >6 mg/day, imipramine, trimipramine [A, H]; amoxapine, desipramine, nortriptyline, protriptyline [A])	Anticholinergic effects including urinary retention ; sedation; orthostatic hypotension, unsteady gait; syndrome of inappropriate antidiuretic hormone (SIADH), so check sodium	Caution in dementia, delirium, falls, lower urinary tract symptoms; avoid in men with benign prostatic hyperplasia (BPH) . Tertiary amines are of special concern in syncope
	First-generation antihistamines (A): brompheniramine, chlorpheniramine, cyproheptadine, dimenhydrinate, diphenhydramine (oral), doxylamine, hydroxyzine, meclizine, promethazine, triprolidine	Highly anticholinergic; clearance — falls with age; tolerance as hypnotics. Cumulative anticholinergic exposure raises falls, delirium and dementia , even in younger adults	
Analgesics	Non-steroidal anti-inflammatory drugs (NSAIDs)	Ulcer, gastrointestinal bleeding, perforation; gastrointestinal risk if age >75 or previous ulcer	Proton pump inhibitor or misoprostol if chronic use (>7 days) or previous ulcer and use is unavoidable. Avoid with corticosteroids or warfarin
	Cyclo-oxygenase 2 (COX-2) inhibitors in heart failure or creatinine clearance (CrCl) <30 mL/min	Worse heart failure and chronic kidney disease; gastrointestinal bleeding in high-risk groups including >75	Avoid in symptomatic heart failure; caution if asymptomatic. Alternatives: paracetamol, serotonin and noradrenaline reuptake inhibitor (SNRI; not duloxetine), topical capsaicin, lidocaine patch
	Tramadol (C)	SIADH, so check sodium; CrCl <30 mL/min raises central nervous system (CNS) effects	In renal impairment avoid the extended-release form and reduce the immediate-release dose
Sex hormones	Oestrogen, oral or transdermal, with or without progestin (A, H)	Breast and endometrial cancer. No cardioprotection, no cognitive protection . If hormone replacement therapy (HRT) is started at ≥60, risks (heart disease, stroke, thrombosis, dementia) outweigh benefits	Vaginal oestrogen is safe and effective for dryness and prevention of urinary tract infection (UTI). After breast cancer that fails non-hormonal therapy, discuss low-dose vaginal oestradiol (<25 mcg twice weekly). Vasomotor symptoms: selective serotonin reuptake inhibitor (SSRI; not paroxetine), SNRI, gabapentin
	Testosterone (A)	Prostate cancer and cardiac events, except in confirmed symptomatic hypogonadism	Beers commentary: "The Use of Testosterone and the Aging Male"

Group	Drug and setting	Concern	Action and alternatives
CNS depressants	Benzodiazepines	Cognitive impairment, unsteady gait, psychomotor impairment, accidents, delirium. May be acceptable for seizures, rapid eye movement sleep disorders, benzodiazepine or alcohol withdrawal, severe generalised anxiety and periprocedural use	Avoid with two or more other CNS-active drugs (falls) and with opioids; reduce other fall-inducing drugs
Anticoagulant	Edoxaban	Avoid if CrCl <15 or >95 mL/min ; no efficacy or safety evidence if <30; reduce dose at CrCl 15–50	Consider warfarin. For atrial fibrillation with CrCl >95: apixaban, dabigatran (if <75 years), rivaroxaban (caution) or warfarin
Antimicrobials	Ciprofloxacin with theophylline or warfarin, or CrCl <30 (A)	Theophylline toxicity; bleeding with warfarin; seizures and confusion in renal impairment	Avoid with theophylline; avoid with warfarin or monitor the international normalised ratio (INR); reduce dose if CrCl <30
	Macrolides other than azithromycin, with warfarin (A)	Bleeding	If unavoidable, monitor INR
	Nitrofurantoin with CrCl <30 (A) or chronic use (A, H)	Pulmonary toxicity, peripheral neuropathy, hepatotoxicity, especially with chronic use	Cohort data suggest it can be effective with minimal risk in moderate renal impairment
	Trimethoprim-sulfamethoxazole (TMP-SMX) with phenytoin or warfarin, or renal insufficiency (especially with an angiotensin-converting enzyme inhibitor [ACEI] or angiotensin receptor blocker [ARB])	Phenytoin toxicity; bleeding; hyperkalaemia ; worse renal function	Avoid with phenytoin; avoid with warfarin or monitor INR; reduce dose at CrCl 15–29; avoid if <15 ; check potassium after 5 days
Antidiuretic	Desmopressin	Hyponatraemia	Treat the cause of nocturia (hyperglycaemia, heart failure, calcium-channel blocker, sodium-glucose co-transporter 2 inhibitor); for BPH consider a 5 α -reductase inhibitor

A, avoid in most older adults (not palliative or hospice care); C, use with caution; H, high-risk medicine in older adults per the Centers for Medicare & Medicaid Services quality measure CMS156v1. Modified from American Geriatrics Society: 2023 updated AGS Beers Criteria®, J Am Geriatr Soc 2023.

- **What it is for:** flags **potentially inappropriate medicines**. Almost every urological prescription touches it.
- **Pattern:** four mechanisms recur. **Anticholinergic burden**, **renal thresholds** (CrCl <30), **warfarin potentiation** (ciprofloxacin, macrolides, TMP-SMX) and **sodium disturbance** (SIADH; desmopressin).
- **Traps:** vaginal oestrogen is not systemic HRT and is endorsed. Nitrofurantoin is listed at CrCl <30 yet usable in moderate impairment. Azithromycin is the exempt macrolide. Duloxetine is excluded as an analgesic alternative and paroxetine as a vasomotor option.
- **Urinary incontinence** — **women are affected more often than men**, including among older adults receiving home care, where roughly two in five report bladder-control problems. **Incontinence becomes more common with age but is neither normal nor inevitable.**

Genitourinary Syndrome of Menopause

- **Genitourinary syndrome of menopause (GSM)** (term coined 2014) affects 27–84% of postmenopausal individuals and is a geriatric-specific risk factor for urgency and urgency urinary incontinence (UUI).
- **More than vaginal atrophy:** besides dyspareunia, pain, dryness and UTI, it causes **urgency, frequency and UI**, and is a common contributor to transient incontinence.
- **Treating atrophy with vaginal oestrogen may relieve some urge incontinence:** urgency or UUI accompanies GSM in about 29%.
- **Hormone deprivation drives overactive bladder (OAB).** Women on oestrogen-deprivation therapy for breast cancer have a **markedly higher OAB hazard**. Men on androgen deprivation therapy (ADT) and men with BPH on 5 α -reductase inhibitors also have more OAB than α -blocker-treated BPH or healthy controls, and the effect grows as ADT approaches 5 years.
- **Before vaginal oestrogen, ask about previous breast or endometrial cancer.** A hormone-sensitive malignancy is **not necessarily a contraindication** to non-systemic vaginal oestrogen, but it **requires consultation with the oncologist**.
- **Urethral caruncle:** a GSM sequela; urethral tissue protrudes, presenting as **spotting on underclothing**. It is usually painless and benign, mimics postmenopausal bleeding, and affects voiding only if large.
- **True postmenopausal bleeding may be an early sign of endometrial or cervical cancer:** work-up finds endometrial cancer in 9%.
- **Caruncle treatment:** topical oestrogen → contraction and involution in most. Excision is rarely necessary but effective.

TREATMENTS FOR GENITOURINARY SYNDROME OF MENOPAUSE

Treatment	Class	Route
Lubricants and moisturisers	Non-hormonal	Vaginal
Oestrogen	Hormonal; also shrinks caruncle	Vaginal (topical)
Dehydroepiandrosterone (DHEA)	Hormonal precursor	Route not specified in the chapter
Hormone replacement therapy	Hormonal	Systemic
Ospemifene	Oestrogen agonist/antagonist	Systemic; route not specified in the chapter

Concept — GSM makes menopause a **lower urinary tract condition**. Loss of sex steroid drives urgency in both sexes, and **local oestrogen** treats bladder, vagina and caruncle together.

Fecal Incontinence

- **Fecal incontinence (FI) is perhaps the most dreaded geriatric syndrome.** Urologists are often involved incidentally although it is not strictly urological.
- **It is under-reported:** fewer than 30% of affected women, and probably fewer men, tell caregivers. **Ask specifically about bowel function and constipation.**
- **Greater dependency predicts more constipation and FI.**
- **Prevalence rises with age and dependency:** 14.4% in the US population, 17% of community-dwelling women over 85 and 43% of women in nursing homes.

Success of Prevention of Geriatric Syndromes

PREVENTIVE INTERVENTIONS IN OLDER ADULTS: WHAT THE EVIDENCE SHOWS

Intervention	Verdict
Calcium and vitamin D	Prevent osteoporosis in both sexes, which reduces fractures
Antihypertensive treatment	Reduces mortality after 80

Intervention	Verdict
Statins	Reduce coronary artery disease in older adults
Physical activity and exercise	Improve mobility, independence and ADLs; possible reduction in dementia (modest evidence, possibly confounded by lifestyle)
Testosterone in hypogonadal men (Testosterone Trials)	No cognitive benefit ; modest improvement in sexual function
Low-dose aspirin in healthy older adults (ASPREE trial)	No gain in disability-free survival; increased all-cause mortality , contrary to long-held belief

Concept — *The value of the syndrome framework is **identification and prevention**. Interventions aimed at a syndrome's drivers work, while inherited assumptions (aspirin, testosterone for cognition) fail when tested.*

Treatment of Incontinence in the Geriatric Patient

- Treatment of UI in older patients **differs in several ways** from treatment in younger patients.
- **Behavioural therapy**: see the American Urological Association (AUA)/Society for Urodynamics and Female Urology (SUFU) stress urinary incontinence guideline (2017), the AUA/SUFU idiopathic OAB guideline (2024) and Chapter 110.

Pharmacotherapy

- **Antimuscarinics improve OAB by 20–70% over placebo, but discontinuation is high.** In older adults, fewer than half still take them at 6 months and about a quarter or fewer at 12 months.
- **Persistence is better** with M3-selective agents, more severe UUI and a well-informed patient. **Cost is a major reason for stopping.**
- **Agents differ** in lipophilicity, molecular size, receptor-subtype selectivity, half-life and metabolism. **The ageing blood–brain barrier is more permeable**, amplifying cognitive effects.
- **Retention is theoretical and uncommon**, but **consider checking post-void residual (PVR)** after starting. **Never exceed the maximum dose.**

ANTIMUSCARINICS VERSUS β 3-AGONISTS IN OLDER ADULTS

	Antimuscarinics	β 3-adrenoceptor agonists
Dry mouth	Strongly increased	Less than placebo
Constipation	Strongly increased	No different from placebo
Cognition	Risk, worsened by blood–brain barrier changes; Beers "avoid" in dementia or delirium	No anticholinergic effect in theory
Cardiovascular	QT prolongation with high-dose tolterodine	Mirabegron: slight rise in blood pressure; interacts with metoprolol
Evidence	Meta-analysis of randomised trials restricted to adults ≥ 65	

AGENT-SPECIFIC POINTS IN OLDER ADULTS

Agent	Class	Distinguishing point
Tropium	Antimuscarinic; quaternary amine (the others are tertiary); M3-selective	Less blood–brain barrier penetration (animal data), so potentially fewer cognitive effects; better persistence
Solifenacin, darifenacin	Antimuscarinic; M3-selective	Better persistence
Tolterodine	Antimuscarinic	High dose $\rightarrow$ QT prolongation $\rightarrow$ symptomatic bradycardia, even torsades de pointes

Agent	Class	Distinguishing point
Oxybutynin, fesoterodine, flavoxate	Antimuscarinic	Listed under Beers "avoid" in dementia or delirium
Mirabegron	β 3-agonist	Metoprolol interaction; use carefully in hypertension

Concept — *In older adults, drug choice turns on **CNS penetration and tolerability, not efficacy**. A quaternary structure (trospium) and a different mechanism (β 3-agonism) are the two ways to avoid anticholinergic burden.*

Incontinence in the Nursing Home Setting

- **Very common in long-term care residents ≥ 65 :** in the US, about half of short-stay and three-quarters of long-stay residents. More than half of residents elsewhere are affected, and **combined urinary and fecal incontinence** is frequent.
- It correlates with a worse **Functional Independence Measure** score and with cognitive impairment, **each of which independently drives long-term-care residence**.
- **UI is a key driver of nursing home admission;** the attributable fraction is larger in men than in women.
- **New UI after admission is common**, suggesting that existing incontinence is unmasked at intake.
- **Reducing catheter-associated UTI (CAUTI) is a priority** in acute and nursing-home care. Standard diagnostic definitions and appropriate catheter use lower catheter use, CAUTI incidence and cost.
- **Colonisation affects up to 98% of indwelling catheters** in older nursing-home residents.
- **Treating asymptomatic bacteriuria (ASB) is harmful.** The Infectious Diseases Society of America (IDSA) advises against it in healthy non-pregnant adults, so **standardised criteria for starting antibiotics are critical:** Loeb's minimum criteria and the National Healthcare Safety Network (NHSN) criteria.

TABLE 120.2 CRITERIA FOR INITIATION OF ANTIBIOTICS FOR URINARY TRACT INFECTION IN RESIDENTS OF LONG-TERM CARE FACILITIES (REBUILT)

Loeb's minimum criteria: resident with catheter	Loeb's minimum criteria: resident without catheter	NHSN criteria
At least one of: (1) fever: single temperature $>100^{\circ}\text{F}$ or $>2^{\circ}\text{F}$ above baseline; (2) new costovertebral tenderness; (3) rigors; (4) acute mental status change	Either: acute dysuria; or temperature $>100^{\circ}\text{F}$ or 2°F above baseline plus at least one new or worsening: urgency, frequency, suprapubic pain, gross haematuria, UI, costovertebral angle tenderness	All three: (1) positive urine culture $\geq 10^5$ colony-forming units; (2) no alternative site of infection; (3) at least one of: fever (single $>100^{\circ}\text{F}$, repeated $>99^{\circ}\text{F}$, or $>2^{\circ}\text{F}$ above baseline), rigors, new hypotension; or leukocytosis ($>14,000/\text{mm}^3$, or left shift $>6\%$ bands or ≥ 1500 bands/ mm^3) plus acute mental status change, acute functional decline, new suprapubic or costovertebral pain or tenderness, purulent discharge around the catheter, or acute testicular, epididymal or prostatic pain, swelling or tenderness

From Loeb et al., Infect Control Hosp Epidemiol 2001; and Stone et al., Infect Control Hosp Epidemiol 2012.

- **Pattern:** the Loeb criteria are **clinical only** and govern **when to start** antibiotics. The catheter set relies on **systemic signs** because catheterised residents cannot report localising symptoms. **NHSN needs culture $\geq 10^5$ plus no alternative source plus clinical signs.**
- **Trap:** a positive culture alone never qualifies. Mental status change counts for the catheterised resident and within NHSN, but not in the non-catheterised set.

Concept — *Where colonisation is the norm, the diagnosis rests on the patient, not the urine. Every criterion set requires systemic or localising signs, because bacteriuria alone has no predictive value.*

Urinary Tract Infections and Asymptomatic Bacteriuria

- **UTI is one of the commonest conditions in older adults**, especially women. More than 50% of women and 12% of men have a UTI in their lifetime.
- **Older adults present atypically**, including with mental status change. However, **a causal link between confusion and UTI is not supported**.
- **Annual UTI incidence in women rises steeply with age**: several-fold higher at 75–84 and ≥85 than at 45–54 (Urologic Diseases in America Project).
- **First task: distinguish symptomatic UTI from OAB and from ASB**.
- **ASB**: bacteria on urine culture, **regardless of pyuria**, without signs or symptoms of UTI. Prevalence reaches 22% in non-institutionalised older women and about 10% in community-dwelling older men.
- ASB raises serum and urinary inflammatory cytokines but **still does not require antibiotics**; outside pregnancy it carries little risk.
- **Immunosuppressed transplant recipients**: evidence is inconclusive, with **no evidence of increased graft failure** from ASB.
- Further reading: AUA core curriculum on recurrent UTI (2023); AUA/Canadian Urological Association (CUA)/SUFU guideline on recurrent uncomplicated UTI in women (2022); Chapter 26.

ASYMPTOMATIC BACTERIURIA: GUIDELINE POSITIONS

Body	Position
US Preventive Services Task Force (USPSTF)	Grade D against screening ; treatment unnecessary and fuels antibiotic resistance
AGS (Choosing Wisely)	Avoiding antibiotics for ASB is one of the original five things to avoid or question
IDSA	Do not treat, except in pregnancy and before genitourinary instrumentation

Concept — *In older adults, bacteriuria is common and confusion is non-specific. The error to avoid is joining the two into a diagnosis of UTI. Only pregnancy and instrumentation justify treating a positive culture without symptoms.*

Prostate Cancer in the Elderly

- Cancers in general affect older people far more than younger people. **Prostate cancer treatment accounts for about 11% of US cancer treatment costs**, possibly because of prostate-specific antigen (PSA) early detection.
- **USPSTF grade D: no PSA screening in men >70**.
- **Estimated remaining life expectancy <10 years → a plan without curative intent**.
- See Chapter 153 and Chapters 158–166.

Bladder Cancer in the Elderly

- **Age is the single greatest risk factor**, and bladder cancer is mainly a disease of older people. It is the fourth commonest cancer and the eighth leading cause of cancer death in US men.
- **AUA guidelines** exist for non-muscle-invasive bladder cancer (NMIBC) and for non-metastatic muscle-invasive bladder cancer (MIBC; AUA/ASCO/ASTRO/SUO), with index case scenarios.
- **Guidelines are often not followed in older patients**. Restaging resection, recommended for T1 intermediate- and high-risk NMIBC, was rarely done in Medicare recipients; it was more frequent with high grade and high stage. **Restaging improved 5-year cancer-specific mortality**.

- **Bacille Calmette–Guérin (BCG) is less effective in older patients** with NMIBC, probably because of age-related decline in immune function.
- **Radical cystectomy can be done safely in older patients** with careful preoperative, intraoperative and postoperative management. **At ≥ 75 , outcomes are comparable to those of younger patients.**
- See Chapters 130–132, 138–139 and 141–144.

Concept — Across urological cancer, the risk in older patients is **undertreatment by age** (*skipped restaging, withheld cystectomy*) as much as **overtreatment** (PSA screening >70). Life expectancy and fitness decide, not age.

END OF LIFE CARE AND UROLOGY

- **Attitudes to palliative and end-of-life care** strongly shape urological health in older adults. **Family members may not share the patient's goals**, so the **goals-of-care conversation comes first**.
- It often starts during an admission, led by the multidisciplinary team. When the urological problem drives the admission, the urologist must be involved.
- **Palliative care is not withdrawal of care**: when cure is no longer possible, treatment moves to palliation. It is no less focused or intense, and **aggressive symptom management is a hallmark**. Urologists play a key role at the end of life in genitourinary disease.
- **Hallmarks of high-quality palliative care**: pain and symptom management; realising personal goals of patient and family; coordination of care.

Resources for Providers

- **AGS**: information for clinicians and families. Families often face this responsibility for the first time and are unsure even of the goals.
- **Urology Care Foundation** (AUA-affiliated): for patients and caregivers.
- Organisations link families with part-time and full-time caregivers. **Medicare, state Medicaid and local county and municipal resources are hard to navigate** for patients and families.

Concept — *Palliation changes the target from cure to symptoms, not the intensity of care, and it starts with agreeing goals, which may differ between patient and family.*

ADDITIONAL RESOURCES

- **Age-Friendly Health Systems: the "4 Ms"**. Mobility, Medication, Mentation, and what Matters to the patient (John A. Hartford Foundation).
- **"Geriatrics at Your Fingertips" (GAYF)**: published annually by the AGS.
- **ACS NSQIP/AGS Best Practice Guidelines**: evidence-based preoperative assessment of older adults across all surgical specialties, in addition to standard work-up.

SUMMARY

- **A comorbidity list classifies disability and dependency poorly**. Geriatric syndromes capture the **negative synergy** unique to older adults, and **treating one syndrome often improves others**. Care is tailored to health, comorbidity and environment.

KEY POINTS (ITEMS NOT COVERED ABOVE)

- Ageing affects the **structure and function of all organ systems**, including the genitourinary system. Health-care delivery must be reimagined to keep pace with growth in the older population.

- Geriatric syndromes also include **pressure ulcers**.
- **UI brings limitation of activity, depression and social isolation**. Individualised treatment improves or resolves many cases. **Transient UI is common**, and treating the underlying condition often resolves it.
- Bladder outlet obstruction, underactive bladder, urinary retention and nocturia are common in older adults.
- **Pelvic organ prolapse is common in older women; age alone is not a reason to withhold surgery**.
- Most genitourinary cancers become more common with age.
- **Sexuality remains important** to quality of life; sexual dysfunction may signal heart disease, diabetes or neurological disease.
- **Urological clinicians help identify elder mistreatment and neglect and are mandatory reporters in most places**.
- **Patient-centred care**: healthy ageing looks different for each person, and care is tailored to the goals of the patient, family and caregivers.